

Exercise C - I

SINGLE CORRECT TYPE QUESTIONS

1. If $A = \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}$ and $I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$, then which one of the following holds for all $n \geq 1$, by the principle of mathematical induction :

यदि $A = \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}$ तथा $I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$ है, तो गणितीय आगमन द्वारा ज्ञात कीजिए कि $n \geq 1$ के लिए निम्न में से कौन सा सत्य है :

(A) $A^n = nA - (n-1)I$

(B) $A^n = 2^{n-1}A - (n-1)I$

(C) $A^n = nA + (n-1)I$

(D) $A^n = 2^{n-1}A + (n-1)I$

Ans. A

Sol. Given, $A = \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}$

Now, $A^2 = \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 2 & 1 \end{bmatrix}$; $A^3 = \begin{bmatrix} 1 & 0 \\ 2 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 3 & 1 \end{bmatrix}$

Similarly, $A^n = \begin{bmatrix} 1 & 0 \\ n & 1 \end{bmatrix}$ can be verified by induction.

Now, go option by option

(b) $\begin{bmatrix} 1 & 0 \\ n & 1 \end{bmatrix} \neq \begin{bmatrix} n & 0 \\ n & n \end{bmatrix} + \begin{bmatrix} n-1 & 0 \\ 0 & n-1 \end{bmatrix}$

(d) $nA - (n-1)I = \begin{bmatrix} n & 0 \\ n & n \end{bmatrix} - \begin{bmatrix} n-1 & 0 \\ 0 & n-1 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ n & 1 \end{bmatrix}$

Hence, option (d) is incorrect.

2. If $P = \begin{bmatrix} \frac{\sqrt{3}}{2} & \frac{1}{2} \\ -\frac{1}{2} & \frac{\sqrt{3}}{2} \end{bmatrix}$, $A = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$ and $Q = PAP^T$ and $x = P^T Q^{2005} P$, then x is equal to :

यदि $P = \begin{bmatrix} \frac{\sqrt{3}}{2} & \frac{1}{2} \\ -\frac{1}{2} & \frac{\sqrt{3}}{2} \end{bmatrix}$, $A = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$ तथा $Q = PAP^T$ तथा $x = P^T Q^{2005} P$ है, तो x का मान है :

(A) $\begin{bmatrix} 1 & 2005 \\ 0 & 1 \end{bmatrix}$

(B) $\begin{bmatrix} 4 + 2005\sqrt{3} & 6015 \\ 2005 & 4 - 2005\sqrt{3} \end{bmatrix}$

(C) $\frac{1}{4} \begin{bmatrix} 2 + \sqrt{3} & 1 \\ -1 & 2 - \sqrt{3} \end{bmatrix}$

(D) $\frac{1}{4} \begin{bmatrix} 2005 & 2 - \sqrt{3} \\ 2 + \sqrt{3} & 2005 \end{bmatrix}$

Ans. A

Sol. $P^T P = I$

$$A = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$$

$$\Rightarrow A^2 = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix} \text{ and so on}$$

$$Q = P A P^T$$

$$Q^2 = (P A P^T) (P A P^T) = P A^2 P^T$$

$$Q^{2005} = P A^{2005} P^T$$

$$x = P^T (P A^{2005} P^T) P \Rightarrow x = A^{2005} = \begin{bmatrix} 1 & 2005 \\ 0 & 1 \end{bmatrix}$$

3. Let $A = \begin{bmatrix} a & b & c \\ p & q & r \\ x & y & z \end{bmatrix}$ and suppose that $\det.(A) = 2$ then the $\det.(B)$ equals, where $B = \begin{bmatrix} 4x & 2a & -p \\ 4y & 2b & -q \\ 4z & 2c & -r \end{bmatrix}$:

माना कि $A = \begin{bmatrix} a & b & c \\ p & q & r \\ x & y & z \end{bmatrix}$ तथा $\det.(A) = 2$ तो $\det.(B)$ का मान कितना होगा जहाँ $B = \begin{bmatrix} 4x & 2a & -p \\ 4y & 2b & -q \\ 4z & 2c & -r \end{bmatrix}$:

(A) $\det(B) = -2$

(B) $\det(B) = -8$

(C) $\det(B) = -16$

(D) $\det(B) = 8$

Ans. C

Sol. $\det(B) = \begin{vmatrix} 4x & 2a & -p \\ 4y & 2b & -q \\ 4z & 2c & -r \end{vmatrix} = (4)(2)(-1) \begin{vmatrix} x & a & p \\ y & b & q \\ z & c & r \end{vmatrix} = -8 \begin{vmatrix} x & y & z \\ a & b & c \\ p & q & r \end{vmatrix} = -8 \begin{vmatrix} a & b & c \\ p & q & r \\ x & y & z \end{vmatrix}$
 $= -8 \times 2 = -16$ Ans.

4. If A is a non singular matrix satisfying $A = AB - BA$, then which one of the following holds true :

यदि A एक व्युत्क्रमणीय आव्यूह है जो $A = AB - BA$, के सन्तुष्ट करती है तब कौन सा कथन सत्य है :

- (A) $\det. B = 0$ (B) $B = 0$ (C) $\det. A = 1$ (D) $\det. (B + I) = \det. (B - I)$

Ans. D

Sol. A is non singular $\det A \neq 0$

Given $AB - BA = A$ hence $AB = A + BA = A(I + B)$

$$\det. A \cdot \det. B = \det. A \cdot \det. (I + B) \quad (\det A \neq 0)$$

$$\det. B = \det. (I + B) \dots (1) \quad (\text{as } A \text{ is non singular})$$

again $AB - A = BA$

$$A(B - I) = BA$$

$$(\det. A) \cdot \det. (B - I) = \det. B \cdot \det. A \quad \Rightarrow \det. (B - I) = \det. (B) \dots (2)$$

from (1) and (2)

$$\det. (B - I) = \det. (B + I) \text{ Ans.}$$

5. $A = \begin{bmatrix} 2 & -1 \\ -7 & 4 \end{bmatrix}$ and $B = \begin{bmatrix} 4 & 1 \\ 7 & 2 \end{bmatrix}$ then $B^T A^T$ is :

- (A) a null matrix (B) an identity matrix
 (C) scalar, but not an identity, matrix (D) such that $\text{tr}(B^T A^T) = 4$

$$A = \begin{bmatrix} 2 & -1 \\ -7 & 4 \end{bmatrix} \text{ तथा } B = \begin{bmatrix} 4 & 1 \\ 7 & 2 \end{bmatrix} \text{ तो } B^T A^T \text{ होगा :}$$

- (A) शून्य आव्यूह (B) इकाई आव्यूह
 (C) सदिश (लेकिन इकाई नहीं) आव्यूह है (D) $\text{tr}(B^T A^T) = 4$

Ans. B

Sol. Compute $(AB)^T$ which is an identity matrix and $(AB)^T = B^T A^T$

6. If $A = \begin{bmatrix} \cos^2 \alpha & \sin \alpha \cos \alpha \\ \sin \alpha \cos \alpha & \sin^2 \alpha \end{bmatrix}$; $B = \begin{bmatrix} \cos^2 \beta & \sin \beta \cos \beta \\ \sin \beta \cos \beta & \sin^2 \beta \end{bmatrix}$ are such that, AB is a null matrix, then which

of the following should necessarily be an odd integral multiple of $\frac{\pi}{2}$:

यदि $A = \begin{bmatrix} \cos^2 \alpha & \sin \alpha \cos \alpha \\ \sin \alpha \cos \alpha & \sin^2 \alpha \end{bmatrix}$; $B = \begin{bmatrix} \cos^2 \beta & \sin \beta \cos \beta \\ \sin \beta \cos \beta & \sin^2 \beta \end{bmatrix}$ इस प्रकार है, कि AB एक शून्य आव्यूह है तब

चरित्र; स्फुटता से इसका बोध; $\frac{\pi}{2}$ का विषम पूर्णांक गुणांक होगा :

- (A) α (B) β (C) $\alpha - \beta$ (D) $\alpha + \beta$

Ans. C

Sol. $AB = \begin{pmatrix} \cos^2 \alpha & \sin \alpha \cos \alpha \\ \sin \alpha \cos \alpha & \sin^2 \alpha \end{pmatrix} \begin{pmatrix} \cos^2 \beta & \sin \beta \cos \beta \\ \sin \beta \cos \beta & \sin^2 \beta \end{pmatrix}$

$$= \begin{pmatrix} \cos^2 \alpha \cos^2 \beta + \sin \alpha \cos \alpha \sin \beta \cos \beta & \cos^2 \alpha \sin \beta \cos \beta + \sin \alpha \cos \alpha \sin^2 \beta \\ \cos^2 \beta \sin \alpha \cos \alpha + \sin^2 \alpha \sin \beta \cos \beta & \sin \alpha \cos \alpha \sin \beta \cos \beta + \sin^2 \alpha \sin^2 \beta \end{pmatrix}$$

$$= \begin{pmatrix} \cos \alpha \cos \beta \cos(\alpha - \beta) & \cos \alpha \sin \beta \cos(\alpha - \beta) \\ \sin \alpha \cos \beta \cos(\alpha - \beta) & \sin \alpha \sin \beta \cos(\alpha - \beta) \end{pmatrix}$$

$\Rightarrow \alpha - \beta$ must be an odd integral multiple of $\pi/2 \Rightarrow (C)$

7. Let A, B, C, D be (not necessarily square) real matrices such that $A^T = BCD$; $B^T = CDA$; $C^T = DAB$ and $D^T = ABC$ for the matrix $S = ABCD$, consider the two statements.

I $S^3 = S$

II $S^2 = S^4$

- (A) II is true but not I (B) I is true but not II
(C) both I and II are true (D) both I and II are false

माना A, B, C, D (जरूरी नहीं है कि वर्ग आव्यूह हो) वास्तविक आव्यूह है तथा $A^T = BCD$; $B^T = CDA$; $C^T = DAB$ तथा $D^T = ABC$ आव्यूह $S = ABCD$, के लिए, दो कथनों पर विचार कीजिए :

I. $S^3 = S$

II. $S^2 = S^4$

- (A) II सत्य है तथा I नहीं (B) I सत्य है तथा II नहीं
(C) I तथा II दोनों सत्य है (D) I तथा II दोनों असत्य है

Ans. C

Sol. $S = ABCD = A(BCD) = AA^T \dots (1)$

$$\begin{aligned} S^3 &= (ABCD)(ABCD)(ABCD) \\ &= (ABC)(DAB)(CDA)(BCD) \\ &= D^T C^T B^T A^T \\ &= (BCD)^T A^T = AA^T \dots (2) \end{aligned}$$

from (1) and (2) $\boxed{S = S^3} \Rightarrow I \text{ is correct}$

multiply both sides by S

$$\boxed{S^2 = S^4} \Rightarrow \text{II is correct} \Rightarrow \text{Both I and II are true Ans.}$$

8. There are two possible values of A in the solution of the matrix equation $\begin{bmatrix} 2A+1 & -5 \\ -4 & A \end{bmatrix}^{-1} \begin{bmatrix} A-5 & B \\ 2A-2 & C \end{bmatrix} = \begin{bmatrix} 14 & D \\ E & F \end{bmatrix}$

where A, B, C, D, E, F are real numbers. The absolute value of the difference of these two solutions, is :

यदि आव्यूह समीकरण $\begin{bmatrix} 2A+1 & -5 \\ -4 & A \end{bmatrix}^{-1} \begin{bmatrix} A-5 & B \\ 2A-2 & C \end{bmatrix} = \begin{bmatrix} 14 & D \\ E & F \end{bmatrix}$ के हल में A के दो सम्भव मान हैं तथा A, B, C, D, E, F

वास्तविक संख्याएँ हैं तो इन दोनों हलों के अन्तर का मापांक क्या होगा :

(A) $\frac{8}{3}$ (B) $\frac{11}{3}$ (C) $\frac{1}{3}$ (D) $\frac{19}{3}$

Ans. D

Sol. If $P = \begin{bmatrix} a & b \\ c & d \end{bmatrix} = \text{adj. } P = \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$

also $A^{-1} = \frac{\text{adj. } A}{\det. A}$

$$\therefore \begin{bmatrix} 2A+1 & -5 \\ -4 & A \end{bmatrix}^{-1} = \frac{\begin{bmatrix} A & 5 \\ 4 & 2A+1 \end{bmatrix}}{2A^2 + A - 20}$$

$$\therefore \frac{1}{2A^2 + A - 20} \begin{bmatrix} A & 5 \\ 4 & 2A+1 \end{bmatrix} \begin{bmatrix} A-5 & B \\ 2A-2 & C \end{bmatrix} = \begin{bmatrix} 14 & D \\ E & F \end{bmatrix}$$

The gives, $\frac{A^2 + 5A - 10}{2A^2 + A - 20} = 14 \Rightarrow A = 3 \text{ or } -\frac{10}{3}$

$$\therefore 3 + \frac{10}{3} = \frac{19}{3} \text{ Ans.}$$

9. Number of real values of λ for which the matrix $A = \begin{bmatrix} \lambda-1 & \lambda & \lambda+1 \\ 2 & -1 & 3 \\ \lambda+3 & \lambda-2 & \lambda+7 \end{bmatrix}$ has no inverse :

(A) 0 (B) 1 (C) 2 (D) infinite

λ के कुल कितने मान होंगे जब आव्यूह $A = \begin{bmatrix} \lambda-1 & \lambda & \lambda+1 \\ 2 & -1 & 3 \\ \lambda+3 & \lambda-2 & \lambda+7 \end{bmatrix}$ व्युत्क्रमणीय नहीं होगा :

- (A) 0 (B) 1 (C) 2 (D) अनन्त

Ans. D

Sol. $|A|=0$ for all $\lambda \in \mathbb{R} \Rightarrow A$ is singular. Hence inverse can not be found for any value of $\lambda \in \mathbb{R} \Rightarrow (D)$

Use $R_2 \rightarrow R_2 - \frac{1}{2}(R_3 - R_1)$

10. Let $A = \begin{bmatrix} l & m & n \\ p & q & r \\ 1 & 1 & 1 \end{bmatrix}$ and $B = A^2$. If $(l-m)^2 + (p-q)^2 = 9$, $(m-n)^2 + (q-r)^2 = 16$, $(n-l)^2 + (r-p)^2 = 25$, then

the value of $\det. B$ equals :

माना $A = \begin{bmatrix} l & m & n \\ p & q & r \\ 1 & 1 & 1 \end{bmatrix}$ तथा $B = A^2$ यदि $(l-m)^2 + (p-q)^2 = 9$, $(m-n)^2 + (q-r)^2 = 16$, $(n-l)^2 + (r-p)^2 = 25$, तो

$\det. B$ का मान होगा :

- (A) 36 (B) 100 (C) 144 (D) 169

Ans. C

Sol. $\det. A$ is twice the area of the triangle with vertices (l, p) , (m, q) , (n, r) with sides 3, 4, 5.

$$\Delta^2 = s(s-a)(s-b)(s-c)$$

$$\Delta^2 = 6(6-3)(6-4)(6-5)$$

$$\Delta^2 = 36 \Rightarrow D = 6$$

Now $\det A = 2\Delta = 12 \Rightarrow \det. B = (\det A)^2 = 4\Delta^2 = 144$ Ans.

11. In a square matrix A of order 3 the elements, a_{ii} 's are the sum of the roots of the equation $x^2 - (a+b)x + ab = 0$; $a_{i,i+1}$'s are the product of the roots, $a_{i,i-1}$'s are all unity and the rest of the elements are all zero. The value of the $\det. (A)$ is equal to :

यदि कोटि 3 के वर्ग आव्यूह A के लिए, अवयवों a_{ii} 's समीकरण में $x^2 - (a+b)x + ab = 0$ के मूलों का योगफल है। $a_{i,i+1}$'s मूलों का गुणनफल है तथा $a_{i,i-1}$'s इकाई है तथा बचे हुए अवयव शून्य है तब $\det. (A)$ का मान बताओ :

- (A) 0 (B) $(a+b)^3$ (C) $a^3 - b^3$ (D) $(a^2 + b^2)(a+b)$

Ans. D

Sol. Given $a_{11} = a_{22} = a_{33} = a+b$

$a_{12} = a_{23} = ab$; $a_{21} = 1 = a_{32}$, all others are zero.

$$\text{Det (A)} = \begin{vmatrix} a+b & ab & 0 \\ 1 & a+b & ab \\ 0 & 1 & a+b \end{vmatrix} = (a^2 + b^2)(a + b)$$

12. Let $A = \begin{bmatrix} 2 & 0 & 7 \\ 0 & 1 & 0 \\ 1 & -2 & 1 \end{bmatrix}$ and $B = \begin{bmatrix} -\alpha & 14\alpha & 7\alpha \\ 0 & 1 & 0 \\ \alpha & -4\alpha & -2\alpha \end{bmatrix}$. If $AB = I$, where I is an identity matrix of order 3 then trace

B has value equal to :

माना $A = \begin{bmatrix} 2 & 0 & 7 \\ 0 & 1 & 0 \\ 1 & -2 & 1 \end{bmatrix}$ तथा $B = \begin{bmatrix} -\alpha & 14\alpha & 7\alpha \\ 0 & 1 & 0 \\ \alpha & -4\alpha & -2\alpha \end{bmatrix}$ यदि $AB = I$, जहाँ I , 3 कोटि का इकाई आव्यूह है तो आव्यूह B का

वृत्त के लंबाई :

(A) 0 (B) $\frac{2}{5}$ (C) $\frac{1}{5}$ (D) 5

Ans. B

Sol. We have $AB = I$, so

$$\begin{bmatrix} 2 & 0 & 7 \\ 0 & 1 & 0 \\ 1 & -2 & 1 \end{bmatrix} \begin{bmatrix} -\alpha & 14\alpha & 7\alpha \\ 0 & 1 & 0 \\ \alpha & -4\alpha & -2\alpha \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} 5\alpha & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 10\alpha - 2 & 5\alpha \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \text{ (Given)}$$

$$\Rightarrow 5\alpha = 1 \Rightarrow \alpha = \frac{1}{5}$$

$$\text{Now, trace (B)} = (-\alpha) + 1 + (-2\alpha) = 1 - 3\alpha = 1 - 3\left(\frac{1}{5}\right) = 1 - \frac{3}{5} = \frac{2}{5}. \text{ Ans.}$$

13. Let D_k is the $k \times k$ matrix with 0's in the main diagonal, unity as the element of 1st row and $(f(k))^{\text{th}}$ column and k for all other entries. If $f(x) = x - \{x\}$ where $\{x\}$ denotes the fractional part function then the value of $\det. (D_2) + \det. (D_3)$ equals :

(A) 32 (B) 34 (C) 36 (D) None

माना D_k , $k \times k$ का एक आव्यूह है, जिसके विकर्ण अवयव '0' है, प्रथम पंक्ति तथा $(f(k))^{th}$ स्तम्भ में 1 है तथा अन्य प्रविष्टियाँ k है।

; $\lim_{x \rightarrow \infty} f(x) = x - \{x\}$ जहाँ पर $\{x\}$ अंश भाग फलन को व्यक्त करता है, तो सारणिक (D_2) तथा सारणिक (D_3) का योग होगा :

- (A) 32 (B) 34 (C) 36 (D) कोई नहीं

Ans. B

Sol. $\text{Det. } (D_2) = \begin{vmatrix} 0 & 1 \\ 2 & 0 \end{vmatrix}$ $(f(2) = 2 \Rightarrow 1^{st} \text{ and } 2^{nd} \text{ column} = 1)$

||ly $\text{Det. } (D_3) = \begin{vmatrix} 0 & 3 & 1 \\ 3 & 0 & 3 \\ 3 & 3 & 0 \end{vmatrix}$ $(f(3) = 3 \Rightarrow a_{13} = 1)$

$\text{Det. } (D_2) = -2$; $\text{Det. } (D_3) = 36$
 $\Rightarrow \text{Det. } (D_2) + \text{Det. } (D_3) = 34$ Ans.

14. Consider a matrix $A(\theta) = \begin{bmatrix} \sin \theta & \cos \theta \\ -\cos \theta & \sin \theta \end{bmatrix}$ then :

- (A) $A(\theta)$ is symmetric (B) $A(\theta)$ is skew symmetric
 (C) $A^{-1}(\theta) = A(\pi - \theta)$ (D) $A^2(\theta) = A\left(\frac{\pi}{2} - 2\theta\right)$

आव्यूह $A(\theta) = \begin{bmatrix} \sin \theta & \cos \theta \\ -\cos \theta & \sin \theta \end{bmatrix}$ पर विचार कीजिए तब :

- (A) $1; \theta$ सममित है (B) $1; \theta$ विषम सममित है
 (C) $A^{-1}(\theta) = A(\pi - \theta)$ (D) $A^2(\theta) = A\left(\frac{\pi}{2} - 2\theta\right)$

Ans. C

Sol. As $A(\theta) = \begin{bmatrix} \sin \theta & \cos \theta \\ -\cos \theta & \sin \theta \end{bmatrix}$

$A(\theta)$ is certainly neither symmetric nor skew symmetric

Further, $A(\pi - \theta) = \begin{bmatrix} \sin \theta & -\cos \theta \\ \cos \theta & \sin \theta \end{bmatrix}$

and $A(\theta) \cdot A(\pi - \theta) = \begin{bmatrix} \sin \theta & \cos \theta \\ -\cos \theta & \sin \theta \end{bmatrix} \begin{bmatrix} \sin \theta & -\cos \theta \\ \cos \theta & \sin \theta \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \Rightarrow A^{-1}(\theta) = A(\pi - \theta)$

$$A^2(\theta) = \begin{bmatrix} \sin \theta & \cos \theta \\ -\cos \theta & \sin \theta \end{bmatrix} \begin{bmatrix} \sin \theta & \cos \theta \\ -\cos \theta & \sin \theta \end{bmatrix} = \begin{bmatrix} -\cos 2\theta & \sin 2\theta \\ -\sin 2\theta & -\cos 2\theta \end{bmatrix} \neq a \left(\frac{\pi}{2} - 2\theta \right)$$

15. Let A and B be 3×3 symmetric matrices such that $X = AB + BA$ and $Y = AB - BA$, then $(XY)^T$ is equal to :
(Note : $(XY)^T$ is the transpose of the matrix XY)

माना A तथा B, 3×3 के सममित आव्यूह है जिनके लिये $X = AB + BA$ तथा $Y = AB - BA$, तो $(XY)^T$ बराबर होगा।
(नोट : $(XY)^T$, XY का परिवर्त आव्यूह है)

- (A) XY (B) YX (C) -XY (D) -YX

Ans. D

Sol. We have $A = A^T$ and $B = B^T$

$$X = AB + BA \Rightarrow X = X^T \quad \text{and} \quad Y = AB - BA \Rightarrow Y = -Y^T$$

$$\text{Now,} \quad (XY)^T = Y^T X^T = -YX. \text{ Ans.}$$

16. If $A = \begin{bmatrix} \alpha & 0 \\ 1 & 1 \end{bmatrix}$ and $B = \begin{bmatrix} 4 & 0 \\ -1 & 1 \end{bmatrix}$, then number of distinct values of α for which $A^2 = B$ equals :

यदि $A = \begin{bmatrix} \alpha & 0 \\ 1 & 1 \end{bmatrix}$ तथा $B = \begin{bmatrix} 4 & 0 \\ -1 & 1 \end{bmatrix}$, तो α के भिन्न मानों की संख्या जिसके के लिये $A^2 = B$ बराबर है :

- (A) 0 (B) 1 (C) 2 (D) 3

Ans. B

Sol. We have

$$A^2 = B \text{ (Given)}$$

$$\Rightarrow \begin{bmatrix} \alpha & 0 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} \alpha & 0 \\ 1 & 1 \end{bmatrix} = \begin{bmatrix} 4 & 0 \\ -1 & 1 \end{bmatrix} \Rightarrow \begin{bmatrix} \alpha^2 & 0 \\ \alpha+1 & 1 \end{bmatrix} = \begin{bmatrix} 4 & 0 \\ -1 & 1 \end{bmatrix}$$

$$\text{So, } \alpha^2 = 4 \text{ and } \alpha + 1 = -1 \Rightarrow \alpha = \pm 2 \text{ and } \alpha = -2$$

$$\therefore \alpha = -2$$

Hence, only one possible value of $\alpha = -2$ exist. Ans.

17. If A is a diagonal matrix of order 3 such that $A^2 = A$, then number of possible matrices A, is ($A \neq O$) :

यदि A, तीन कोटि का विकर्ण आव्यूह $A^2 = A$ है, तो संभव आव्यूहों A की संख्या होगी ($A \neq O$) :

- (A) 2 (B) 3 (C) 8 (D) 7

Ans. D

Sol. $A^2 = A$

$\Rightarrow d_1, d_2, \dots, d_n$ be the diagonal matrix of A, then

$$A^2 = A \Rightarrow d_1^2 = d_1, d_2^2 = d_2, \dots, d_n^2 = d_n$$

Now, $d_i^2 = d_i \Rightarrow$ either $d_i = 0$ or $d_i = 1$

Number of ways in which diagonal elements can be filled up is 2^3 , but all diagonal elements cannot be zero.

$\therefore$ Number of required matrices is $2^3 - 1 = 7$. Ans.

18. If $\omega \neq 1$ is the complex cube root of unity and matrix $H = \begin{bmatrix} \omega & 0 \\ 0 & \omega \end{bmatrix}$, then H^{70} is equal to : [AIEEE-2011]

यदि $\omega \neq 1$ इकाई का एक सम्मिश्र घन मूल है तथा आव्यूह $H = \begin{bmatrix} \omega & 0 \\ 0 & \omega \end{bmatrix}$ हैं, तो H^{70} बराबर है : [AIEEE-2011]

- (A) H (B) 0 (C) $-H$ (D) H_2

Ans. A

Sol. Since H is a diagonal matrix. We know that product of two diagonal matrix is always a diagonal matrix.

$$\begin{aligned} \text{So } H^{70} &= \begin{bmatrix} \omega & 0 \\ 0 & \omega \end{bmatrix} \begin{bmatrix} \omega & 0 \\ 0 & \omega \end{bmatrix} \dots \begin{bmatrix} \omega & 0 \\ 0 & \omega \end{bmatrix} \text{ 70 times} \\ &= \begin{bmatrix} \omega^{70} & 0 \\ 0 & \omega^{70} \end{bmatrix} = \begin{bmatrix} \omega & 0 \\ 0 & \omega \end{bmatrix} = H \end{aligned}$$

19. Let $\omega \neq 1$ be a cube root of unity and S be the set of all non-singular matrices of the form $\begin{bmatrix} 1 & a & b \\ \omega & 1 & c \\ \omega^2 & \omega & 1 \end{bmatrix}$, where

each of a, b and c is either ω or ω^2 . Then the number of distinct matrices in the set S is : [JEE-2011]

माना $\omega \neq 1$ इकाई का घनमूल है तथा S वह सभी व्युत्क्रमणीय आव्यूह $\begin{bmatrix} 1 & a & b \\ \omega & 1 & c \\ \omega^2 & \omega & 1 \end{bmatrix}$ का सम्मुचय है, जहाँ a, b तथा c ω या ω^2 में

है। S में सम्भव भिन्न आव्यूहों की संख्या होगी :

[JEE-2011]

- (A) 2 (B) 6 (C) 4 (D) 8

Ans. A

Sol. $\begin{vmatrix} 1 & a & b \\ \omega & 1 & c \\ \omega^2 & \omega & 1 \end{vmatrix}$

$$= 1 - c\omega - a(\omega - \omega^2 c) = (1 - c\omega) - a\omega(1 - c\omega) = (1 - c\omega)(1 - a\omega)$$

for non singular matrix

$$c \neq \frac{1}{\omega} \text{ and } a \neq \frac{1}{\omega}$$

$$\Rightarrow c \neq \omega^2, \quad a \neq \omega^2$$

$$\Rightarrow a \text{ and } c \text{ must be } \omega \text{ and } b \text{ can be } \omega \text{ or } \omega^2$$

$$\therefore \text{ total matrices} = 2$$

ONE OR MORE THAN ONE CORRECT TYPE QUESTIONS

20. If A and B are different matrices satisfying $A^3 = B^3$ and $A^2B = B^2A$, then :

- (A) $\det(A^2 + B^2)$ must be zero
 (B) $\det(A - B)$ must be zero
 (C) $\det(A^2 + B^2)$ as well as $\det(A - B)$ must be zero
 (D) At least one of $\det(A^2 + B^2)$ or $\det(A - B)$ must be zero

यदि A एवं B भिन्न-भिन्न आव्यूह समीकरणों $A^3 = B^3$ तथा $A^2B = B^2A$ को संतुष्ट करते हैं, तो :

- (A) $\det(A^2 + B^2) = 0$
 (B) $\det(A - B) = 0$
 (C) कमज $\det(A^2 + B^2)$ तथा $\det(A - B)$ दोनों शून्य होगा
 (D) कमज $\det(A^2 + B^2)$ तथा $\det(A - B)$ में कम से कम एक शून्य अवश्य होगा

Ans. AD

Sol. $A^3 = B^3$ (1)

and $A^2B = B^2A$ (2)

(1) - (2) gives, $A^3 - A^2B = B^3 - B^2A$

$$A^2(A - B) = -B^2(A - B) \Rightarrow (A^2 + B^2)(A - B) = 0$$

$$\det(A^2 + B^2)(A - B) = 0$$

$$\det(A^2 + B^2) \cdot \det(A - B) = 0 \Rightarrow \text{(D)}$$

21. Given the matrices A and B as $A = \begin{bmatrix} 1 & -1 \\ 4 & -1 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & -1 \\ 2 & -2 \end{bmatrix}$. The two matrices X and Y are such that $XA = B$ and $AY = B$ then which of the following hold(s) true?

दिये गये आव्यूह A तथा B जहाँ $A = \begin{bmatrix} 1 & -1 \\ 4 & -1 \end{bmatrix}$ और $B = \begin{bmatrix} 1 & -1 \\ 2 & -2 \end{bmatrix}$ दो आव्यूह X और Y इस प्रकार हैं कि $XA = B$ और $AY = B$

तो निम्न में से कौन सा कथन सत्य है?

$$(A) X = \frac{1}{3} \begin{bmatrix} 1 & -1 \\ -2 & 2 \end{bmatrix} \quad (B) Y = \frac{1}{3} \begin{bmatrix} 3 & 0 \\ 4 & 0 \end{bmatrix} \quad (C) \det. X = \det. Y \quad (D) 3(X + Y) = \begin{bmatrix} 4 & -1 \\ 4 & 2 \end{bmatrix}$$

Ans. CD

Sol. Note that A is non singular but B is singular hence only A^{-1} exists

$$\text{Now} \quad XA = B \quad \Rightarrow \quad X = BA^{-1} \quad \underline{\hspace{2cm}} (1)$$

$$\text{and } AY = B \quad \Rightarrow \quad Y = A^{-1}B \quad \underline{\hspace{2cm}} (2)$$

$$\text{Also} \quad A^{-1} = \frac{1}{3} \begin{bmatrix} -1 & 1 \\ -4 & 1 \end{bmatrix}$$

now verify

22. If there are three square matrix A, B, C of same order satisfying the equation $A^2 = A^{-1}$ and let $B = A^{2^n}$ and

$C = A^{2^{(n-2)}}$ then which of the following statements are true?

$$(A) \det. (B - C) = 0 \quad (B) (B + C)(B - C) = 0$$

$$(C) B \text{ must be equal to } C \quad (D) \text{ None}$$

यदि तीन वर्ग आव्यूह A, B, C समीकरण $A^2 = A^{-1}$ एवं माना $B = A^{2^n}$ तथा $C = A^{2^{(n-2)}}$ है, तो?

$$(A) \det. (B - C) = 0 \quad (B) (B + C)(B - C) = 0$$

$$(C) \text{ आव्यूह } B, C \text{ के समान है} \quad (D) \text{ कोई नहीं}$$

Ans. ABC

$$\text{Sol.} \quad B = A^{2^n} = A^{2 \cdot 2^{n-1}} = (A^2)^{2^{n-1}} = (A^{-1})^{2^{n-1}} = \left(A^{2^{n-1}}\right)^{-1} = \left(A^{2 \cdot 2^{n-2}}\right)^{-1} = \left((A^2)^{2^{n-2}}\right)^{-1}$$

$$= \left((A^{-1})^{-1}\right)^{2^{n-2}} = A^{2^{(n-2)}}$$

so $B = C$

$$(B - C) = \text{null} \quad \Rightarrow \quad (B + C)(B - C) = 0 \Rightarrow \det. (B - C) = 0$$

$$\Rightarrow A, B, C \text{ are answers}$$